

SGA 5 — Exposé I

Dualizing Complexes

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Introduction

This exposé is devoted to the biduality theorem in étale cohomology. The theory developed here and in SGA A XVIII (SGA A = SGA 4), for the étale topology and constructible torsion sheaves on preschemes, is formally very analogous to the one developed in HARTSHORNE [H] for the Zariski topology and coherent sheaves: the latter served, to a very large extent, as its model. But, whereas in the case of “continuous” coefficients the existence of dualizing complexes presents no difficulties, in the case of “discrete” coefficients, on the other hand, it is much more delicate to prove, and it seems to require resolution of singularities in an essential way. See however Deligne’s biduality theorem (SGA 4^{1/2}, Finiteness Theorem 4.3.).

Paragraph 1 contains the definition of dualizing complexes and the formal properties that follow from it, notably the interpretation of dualizing functors as “exchangers of functors.”

In paragraph 2, one shows that, on a connected locally noetherian prescheme, a dualizing complex is determined uniquely, up to a translation of degrees and tensoring by an invertible sheaf.

In paragraph 3, one comes to the central theorem of the exposé (3.4.1.), which asserts that, on a “good” regular prescheme X , for example on an excellent noetherian regular prescheme of characteristic zero¹, the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing.

Paragraph 4 gives the applications of the theory to the local duality theorem. The latter expresses that, on a “good” prescheme X endowed with a closed point x whose residue field is separably closed, and for a constructible sheaf F , one has a perfect pairing between $\underline{H}_x^i(F')$ and $\underline{H}_x^{-i}(F)$, where F' is the dual of F , i.e. $F' = R\underline{\mathrm{Hom}}(F, K_X)$, where K_X is a dualizing complex.

Finally, in paragraph 5, one proves directly that on a regular prescheme of dimension 1 the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing (n prime to the residual characteristics).

1 Definition and formal properties of dualizing complexes

We fix a coefficient ring $A = \mathbb{Z}/n\mathbb{Z}$. If X is a prescheme, we denote by A_X the constant sheaf on $X_{\text{ét}}$ with value A , and by $D(X)$ the derived category of the category of A_X -modules. We denote, on the other hand, by $D_c(X)$ the full triangulated subcategory of $D(X)$ formed by the complexes F such that $H^i(F)$ is a constructible A_X -module for every i . We put $D_c^*(X) = D^*(X) \cap D_c(X)$, where $*$ = $\emptyset$, $+$, $-$, or b .

¹This last restriction will become unnecessary once one has resolution of singularities à la Hironaka for these preschemes, and the “purity theorem.”

Definition 1.1

An A_X -module I is said to be *quasi-injective* if one has

$$\underline{\mathrm{Ext}}^i(F, I) = 0$$

for every *constructible* A_X -module F and every $i > 0$.

Remarks 1.2

- a) Contrary to what happens in the case of “continuous” coefficients (cf. [H] II.7.17), a quasi-injective sheaf is not injective in general. For example, let $X = \mathrm{Spec}(\mathbb{R})$ and $A = \mathbb{Z}/2\mathbb{Z}$: the sheaf A_X is quasi-injective, but is not injective, nor even of finite injective dimension, since $H^*(X; A_X) = H^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}/2\mathbb{Z})$ is a polynomial algebra on one generator of dimension 1.
- b) Let I be a quasi-injective A_X -module. It is false in general that the relation $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > 0$, valid for constructible F , is valid for every F . Indeed, take $A = \mathbb{F}_\ell$. Let k be a field, $\bar{k}$ a separable closure of k , and $f : \bar{x} = \mathrm{Spec}(\bar{k}) \rightarrow X = \mathrm{Spec}(k)$ the canonical morphism, and suppose that, for every connected étale covering $X' \rightarrow X$, there exists a connected étale covering $X'' \rightarrow X'$ such that $[X'' : X']$ is divisible by ℓ (take, for example, $k = \mathbb{Q}$). Consider the exact sequence

$$(*) \quad 0 \longrightarrow A_X \xrightarrow{\varepsilon} f_* f^* A_X \longrightarrow F \longrightarrow 0$$

defined by the adjunction arrow ε . In view of the hypothesis, the section of $\mathrm{Ext}^1(F, A_X)$ defined by $(*)$ does not vanish above any U étale over X ; hence $\underline{\mathrm{Ext}}^1(F, A_X)_{\bar{x}} \neq 0$. But one checks trivially that A_X is quasi-injective.

Proposition 1.3 (cf. [H] 1.7.6)

Let X be a prescheme and $N \in \mathbb{Z}$. The following conditions on $K \in \mathrm{ob} D^+(X)$ are equivalent:

- (i) K is isomorphic, in $D(X)$, to a bounded complex I such that I^i is quasi-injective for every i and zero for $i > N$;
- (ii) for every $F \in \mathrm{ob} D_c^b(X)$ such that $H^i(F) = 0$ for $i < 0$, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$;
- (iii) for every constructible A_X -module F , one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$.

Proof. (i) $\Rightarrow$ (ii). Let $F \in \mathrm{ob} D_c^b(X)$ be such that $H^i(F) = 0$ for $i < 0$, and let $I \in \mathrm{ob} D^+(X)$ be bounded and such that I^i is quasi-injective for every i and zero for $i > N$; we show that $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > N$. Arguing by induction on the number of indices i such that $H^i \neq 0$, one is reduced, by dévissage, to the case where $N = 0$ and I is concentrated in degree 0. One then has a biregular spectral sequence

$$E_2^{pq} = \underline{\mathrm{Ext}}^p(H^{-q}(F), I) \Rightarrow \underline{\mathrm{Ext}}^*(F, I).$$

Since I is quasi-injective, $E_2^{pq} = 0$ for $p > 0$ and every q , and hence one has $E_2^{0i} = \underline{\mathrm{Hom}}(H^{-i}(F), I) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(F, I)$, which gives the conclusion.

(ii) $\Rightarrow$ (iii) is trivial.

(iii) $\Rightarrow$ (i) We may suppose that K is bounded below and has injective components. The hypothesis, applied to $F = A_X$, implies that $H^i(K) = 0$ for $i > N$. We may therefore replace K by an isomorphic complex K' , bounded, and such that K'^i is injective for $i < N$ and zero for $i > N$. It is then observed that $K'^{!N}$ is quasi-injective, which completes the proof.

Remark 1.3.1

The writer does not know whether (1.3) is still true when, in condition (ii), the hypothesis “ $F \in \text{ob } D_c^b(X)$ ” is replaced by “ $F \in \text{ob } D_c^+(X)$ ”.

Definition 1.4

The *quasi-injective dimension* of K , denoted $\dim . q. inj(K)$, is the lower bound (in $\overline{\mathbb{R}}$) of the integers N satisfying the equivalent conditions of (1.3).

If K is of finite quasi-injective dimension, the functor $R \underline{\text{Hom}}(_, K) : D(X) \rightarrow D(X)$ induces a functor $D_c^b(X) \rightarrow D^b(X)$.

We shall see below that if X is smooth of dimension d over a field k and n is prime to the characteristic of k , then $\dim . q. inj(A_X) = 2d$.

The example of (1.2 a)) shows that a complex of finite quasi-injective dimension is not necessarily of finite injective dimension. Nevertheless one has:

Proposition 1.5

Let X be a locally noetherian prescheme and $K \in \text{ob } D^+(X)$. Suppose that there exists an integer N such that $\text{cd}_{\mathbf{n}}(X) \leq N$, where $\mathbf{n}$ is the set of prime divisors of n (notation of SGAA X 1). Then, for $N' \in \mathbb{Z}$, one has

$$\dim . q. inj(K) \leq N' \implies \dim . inj(K) \leq N + N'.$$

We shall use, in the proof, the following

Lemma 1.5.1

Let C be an abelian category satisfying (AB5) and possessing a family of generators $(U_i)_{i \in I}$. Let $K \in \text{ob } D^+(C)$ and $N \in \mathbb{Z}$. The following conditions are equivalent:

- (i) $\dim . inj(K) \leq N$;
- (ii) for every $i \in I$ and every quotient F of U_i , one has

$$\text{Ext}^n(F, K) = 0 \quad \text{for } n > N.$$

Proof. It suffices to prove (ii) $\Rightarrow$ (i). By a well-known argument (cf. for example [H] 1.7.6), one is reduced to showing that if $K' \in \text{ob}(C)$ is such that $\text{Ext}^1(F, K') = 0$ every time F is a quotient of a U_i , then K' is injective. In other words, one has to prove that, if every morphism from a subobject

V of a U_i into K' extends to U_i , then K' is injective. But this follows immediately from the proof of Lemma 1, p. 136 of (Tohoku).

Proof of (1.5). Suppose $\dim .q.inj(K) \leq N'$ and prove $\dim .inj(K) \leq N + N'$. Since the category of A_X -modules admits as generators the $A_{U,X}$ where U is étale of finite type over X , it suffices, by (1.5.1), to prove that $\text{Ext}^i(X; F, K) = 0$ for $i > N + N'$ when F is a quotient of such an $A_{U,X}$, hence constructible (SGAA IX 2.9). Consider the spectral sequence

$$E_2^{pq} = H^p(X; \underline{\text{Ext}}^q(F, K)) \Rightarrow \text{Ext}^*(X; F, K).$$

The hypothesis $\text{cd}_n(X) \leq N$ (resp. $\dim .q.inj(K) \leq N'$) implies $E_2^{pq} = 0$ for $p > N$ (resp. $q > N'$). Hence $E_2^{pq} = 0$ for $p + q > N + N'$, whence the conclusion.

Remark 1.5.2

The hypothesis of (1.5) is verified (with $N = 2d$) if X is a prescheme of finite type and dimension d over a separably closed field k , with n prime to $\text{char}(k)$ (SGAA X 4).

Proposition 1.6

Let $f : X \rightarrow Y$ be a separated quasi-finite morphism of noetherian preschemes, and let $K \in \text{ob } D^+(Y)$. Then, for $N \in \mathbb{Z}$, one has

$$\dim .q.inj(K) \leq N \quad \implies \quad \dim .q.inj(R^!f(K)) \leq N.$$

Proof. According to the “Main theorem” (EGA IV 8.12.6), there exists a factorization of f as $f = uf'$, where f' is an open immersion and u is a finite morphism. We have $R^!f = R^!f' R^!u$, which reduces us to examining separately the case of an open immersion and that of a finite morphism. If f is an open immersion, the assertion is trivial (since every constructible sheaf on X is then induced by a constructible sheaf on Y). Suppose therefore that f is a finite morphism, and let F be a constructible A_X -module. We then have the duality isomorphisms (SGAA XVIII 3.1.9.6)

$$f_* \underline{\text{Ext}}^i(F, R^!f(K)) \xrightarrow{\sim} \underline{\text{Ext}}^i(f_* F, K).$$

Since $f_*(F)$ is constructible (SGAA IX 2.14), it follows that, if $\dim .q.inj(K) \leq N$, then $f_* \underline{\text{Ext}}^i(F, R^!f(K)) = 0$ for $i > N$, hence $\underline{\text{Ext}}^i(F, R^!f(K)) = 0$ for $i > N$ (SGAA VIII 5.5), as required.

Let now X be a prescheme, and let $K \in \text{ob } D^+(X)$. Denote by $\underline{D}_K$ the functor $R \underline{\text{Hom}}(-, K)$. We shall define, for $F \in \text{ob } D(X)$, a functorial homomorphism $F \rightarrow \underline{D}_K \underline{D}_K F$ (cf. [H] V.1.2). The construction which follows would be valid more generally on a ringed topos. We may first suppose that K is formed of injectives, which allows us to “remove the R ”. The identity morphism $\underline{\text{Hom}}(F, K) \rightarrow \underline{\text{Hom}}(F, K)$ defines, by the formula dear to Cartan, a homomorphism $F \otimes \underline{\text{Hom}}(F, K) \rightarrow K$, whence, by a second application of the said formula, a homomorphism $F \rightarrow \underline{\text{Hom}}(\underline{\text{Hom}}(F, K), K)$, which is the announced homomorphism, and which we baptize the canonical homomorphism. We say that the pair (F, K) is *bidualizing*, or *satisfies biduality*, or again that F is *reflexive* relative to K , if the canonical homomorphism $F \rightarrow \underline{D}_K \underline{D}_K F$ is an isomorphism.

From now on, unless expressly mentioned otherwise, all preschemes considered will be assumed lo

Definition 1.7

Let X be a prescheme. A complex $K \in \text{ob } D^+(X)$ is said to be *dualizing* if the following conditions are satisfied:

- (i) K is of finite quasi-injective dimension;
- (ii) for every $F \in \text{ob } D_c^-(X)$, one has $\underline{D}_K(F) \in \text{ob } D_c^+(X)$;
- (iii) every $F \in \text{ob } D_c^b(X)$ is reflexive relative to K .

Remark 1.8

- a) By a standard dévissage (truncating F and using the fact that $D_c(X)$ is triangulated), one sees that condition (ii) is equivalent to
(ii bis): for every constructible A_X -module F and every $i \in \mathbb{Z}$, $\underline{\text{Ext}}^i(F, K)$ is constructible.
- b) Condition (ii bis) implies $K \in \text{ob } D_c^+(X)$. Contrary to the example of coherent sheaves (EGA 0_{III} 12.3.3), the converse is perhaps not automatically verified. It is however verified (see no. 3) in the “good” cases, for example when one has on X resolution of singularities and purity (in particular if X is an excellent prescheme of characteristic zero).
- c) By dévissage on F , one sees that condition (iii) is equivalent to (iii bis): every constructible A_X -module F is reflexive relative to K .
- d) If $K \in \text{ob } D^+(X)$ satisfies (i) and (ii) (resp. is dualizing), the functor $\underline{D}_K$ induces a functor (resp. an equivalence of categories) $(D_c^b(X))^\circ \rightarrow D_c^b(X)$.
- e) If $K \in \text{ob } D^+(X)$ is of finite *injective* dimension (cf. 1.5.2) and satisfies (ii), then, for every $F \in \text{ob } D_c(X)$, one has $\underline{D}_K(F) \in D_c(X)$, and $\underline{D}_K$ induces a functor $(D_c^+(X))^\circ \rightarrow D_c^-(X)$.
If moreover K is dualizing, every $F \in \text{ob } D_c(X)$ is reflexive relative to K , and $\underline{D}_K$ induces equivalences of categories

$$(D_c(X))^\circ \xrightarrow{\sim} D_c(X), \quad (D_c^-(X))^\circ \xrightarrow{\sim} D_c^+(X), \quad (D_c^+(X))^\circ \xrightarrow{\sim} D_c^-(X).$$

It is not known whether this conclusion remains valid without the restriction that K be of finite injective dimension.

We now turn to the “exchange formulas.” We shall not return to the notion of the tor-dimension of a complex, see SGA A, Exposé XVII, 4.1.9, and also SGA 6, I, 5. Retain only the following point: if X is a prescheme and $G \in \text{ob } D^-(X)$, the condition $\underline{\text{Tor}}_i(F, G) = 0$ for $i > N$ and all A_X -modules F is equivalent to the same condition with F constructible, by SGA A, Exposé IX, 2.9 and the commutation of tensor product with filtered inductive limits. Thus there is no need to introduce a notion of “quasi-tor-dimension.”

Lemma 1.9, cf. [H] II.4.3.

- a) Let X be a prescheme and let $F, G \in \text{ob } D_c^-(X)$. One has $F \otimes^L G \in \text{ob } D_c(X)$ if $F \in \text{ob } D_c^-(X)$, or if G has finite tor-dimension.

b) Let $f : X \rightarrow Y$ be a morphism of preschemes and let $G \in \text{ob } D_c^-(Y)$. Then $f^*(G) \in \text{ob } D_c(X)$.

Proof. Part (b) follows trivially from SGA A, Exposé IX, 2.4. We prove (a). Either hypothesis on (F, G) implies that the spectral sequence

$$E_2^{pq} = \sum_{q_1+q_2=q} \underline{\text{Tor}}_{-p}(\mathbf{H}^{q_1}(F), \mathbf{H}^{q_2}(G)) \Rightarrow \mathbf{H}^*(F \otimes^{\mathbf{L}} G)$$

is biregular. Hence one is reduced to the case where F and G are concentrated in degree 0, that is, are constructible A_X -modules. The question is local, so one may suppose X noetherian. Then F admits, by SGA A, Exposé IX, 2.7, a left resolution F' in which the F'^i are of the form $A_{U,X}$ with U étale and of finite type over X . We have

$$F \otimes^{\mathbf{L}} G \xleftarrow{\sim} F' \otimes G,$$

and the result follows since the $A_{U,X} \otimes G$ are constructible. $\square$

Notation 1.10. If X is a prescheme, we denote by $D^b(X)_{\text{torf}}$, respectively $D^b(X)_{\text{qinj}}$, the full subcategory of $D(X)$ consisting of objects of finite tor-dimension, respectively of finite quasi-injective dimension.

Proposition 1.11, cf. [H] V.2.6. Let X be a prescheme and $K \in \text{ob } D^+(X)$.

a) For $F, G \in \text{ob } D^-(X)$, respectively for $F \in \text{ob } D(X)$ and $G \in \text{ob } D^b(X)_{\text{torf}}$, there is a functorial isomorphism

$$\underline{D}_K(F \otimes^{\mathbf{L}} G) \xrightarrow{\sim} \mathbf{R} \underline{\text{Hom}}(F, \underline{D}_K(G)).$$

b) If K satisfies conditions (i) and (ii) of Definition 1.7, then

$$G \in \text{ob } D_c^b(X)_{\text{torf}} \implies \underline{D}_K(G) \in \text{ob } D_c^b(X)_{\text{qinj}}$$

and $\underline{D}_K(G)$ satisfies condition 1.7(ii).

c) If K is dualizing, then for $F \in \text{ob } D_c^-(X)$ and $G \in \text{ob } D_c^b(X)$, respectively for $F \in \text{ob } D(X)$ and $G \in \text{ob } D_c^b(X)$ with $\underline{D}_K(G) \in \text{ob } D_c^b(X)_{\text{torf}}$, there is a functorial isomorphism

$$\underline{D}_K(F \otimes^{\mathbf{L}} \underline{D}_K(G)) \xrightarrow{\sim} \mathbf{R} \underline{\text{Hom}}(F, G).$$

Moreover,

$$\underline{D}_K(G) \in \text{ob } D_c^b(X)_{\text{torf}} \implies G \in \text{ob } D_c^b(X)_{\text{qinj}},$$

and G satisfies condition 1.7(ii).

d) If K is dualizing and has finite injective dimension, then the implications in (b) and (c) are equivalences.

Proof. Part (a) is precisely Cartan's adjunction formula, SGA 6, I, 7.4.

For (b), let $G \in \text{ob } D_c^b(X)_{\text{torf}}$. If F is a variable constructible A_X -module, then the cohomology objects of $F \otimes^{\mathbf{L}} G$ are constructible, by Lemma 1.9, and vanish outside two fixed bounds independent

of F . The same therefore holds for the cohomology objects of $\underline{D}_K(F \otimes^L G)$, and (a) gives the assertion.

For (c), the first assertion follows from (a) applied to $\underline{D}_K(G)$ in place of G , together with the reflexivity of G . The second follows by a similar argument to (b).

For (d), prove for instance the converse implication of (b). Suppose that $\underline{D}_K(G) \in \text{ob } D_c^b(X)_{\text{qinj}}$ and satisfies condition 1.7(ii). Then, as F varies through constructible modules, the cohomology objects of $\underline{D}_K(F \otimes^L G)$ are constructible and vanish outside two fixed bounds. The same is consequently true for the cohomology objects of $\underline{D}_K \underline{D}_K(F \otimes^L G)$; but

$$F \otimes^L G \xrightarrow{\sim} \underline{D}_K \underline{D}_K(F \otimes^L G)$$

because K is dualizing and of finite injective dimension, cf. 1.8(d). Hence G has finite tor-dimension. The converse implication in (c) is proved analogously. $\square$

Remarks 1.11.1.

- a) The editor does not know whether assertion (d) of Proposition 1.11 is valid under the sole hypothesis that K be dualizing.
- b) One expresses the preceding proposition pictorially by saying that the dualizing functor $\underline{D}_K$ exchanges the functors $\otimes^L$ and $\text{R}\underline{\text{Hom}}$, as well as the notions of finite tor-dimension and finite quasi-injective dimension. The exchange is, however, fully satisfactory only when K has finite injective dimension.

Proposition 1.12. Let Y be a noetherian prescheme, let $f : X \rightarrow Y$ be a morphism of finite type, and let $K \in \text{ob } D^+(Y)$. Suppose that n is prime to the residual characteristics of Y , and put

$$K_X = \text{R}^! f(K_Y), \quad \underline{D}_X = \underline{D}_{K_X}, \quad \underline{D}_Y = \underline{D}_{K_Y},$$

in the notation of SGA A, Exposé XVIII, 3.1.

- a) For $F \in \text{ob } D^-(X)$ there is a functorial isomorphism

$$\text{R}f_* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \text{R}f_*(F). \quad (\text{i})$$

If, moreover, K_X and K_Y are dualizing,² then for $F \in \text{ob } D_c^b(X)$ there is a functorial isomorphism

$$\text{R}f_! \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \text{R}f_*(F). \quad (\text{ii})$$

- b) For $F \in D_c^-(Y)$ there is a functorial isomorphism³

$$\text{R}^! f \underline{D}_Y(F) \xrightarrow{\sim} \underline{D}_X f^*(F). \quad (\text{i})$$

If, moreover, K_X and K_Y are dualizing, then for $F \in D_c^b(Y)$ there is a functorial isomorphism

$$f^* \underline{D}_Y F \xrightarrow{\sim} \underline{D}_X \text{R}^! f F. \quad (\text{ii})$$

²We shall see below that under fairly general conditions K_Y dualizing implies K_X dualizing.

³If the functor $\text{R}^! f$ has finite cohomological dimension, as for example when Y is of finite type over a field (SGA A, Exposé X, 4.3 and Exposé XVIII, 3.1.7), the isomorphism (b)(i) is valid for $F \in \text{ob } D_c(Y)$.

Proof. The isomorphism (a)(i) is just the duality isomorphism of SGA A, Exposé XVIII, 3.1.9.6.

The isomorphism (a)(ii) is obtained by applying $\underline{D}_Y$ to both sides of (a)(i), written with argument $\underline{D}_X F$, and using the finiteness theorem SGA A, Exposé XVII, 5.3.6, which asserts that $Rf_*(\underline{D}_X(F)) \in \text{ob } D_c(Y)$.

The isomorphism (b)(i) is the induction formula of SGA A, Exposé XVIII, 3.1.12.2.

The isomorphism (b)(ii) is obtained by applying $\underline{D}_X$ to both sides of (b)(i), written with argument $\underline{D}_Y F$, and using 1.9(b). $\square$

Scholium 1.12.1. In figurative language, the dualizing functors $\underline{D}_X$ and $\underline{D}_Y$ exchange the usual and unusual direct images, Rf_* and $Rf_!$, and also the usual and unusual inverse images, f^* and $R^!f$.

Corollary 1.13. Under the hypotheses of Proposition 1.12, suppose in addition that f is proper, and let $F \in \text{ob } D^-(X)$. If (F, K_X) is bidualizing, then (Rf_*F, K_Y) is bidualizing. Conversely, if f is finite and (Rf_*F, K_Y) is bidualizing, then (F, K_X) is bidualizing.

Proof. Suppose (F, K_X) is bidualizing. Applying Rf_* to the canonical isomorphism $F \xrightarrow{\sim} \underline{D}_X^2 F$, one obtains

$$\begin{aligned} Rf_*F &\xrightarrow{\sim} Rf_*\underline{D}_X\underline{D}_X F \\ &\xrightarrow{\sim} \underline{D}_Y Rf_*\underline{D}_X F && \text{by (a)(i)} \\ &\xrightarrow{\sim} \underline{D}_Y \underline{D}_Y Rf_*F && \text{by (a)(i).} \end{aligned}$$

The courageous reader will convince himself that the isomorphism obtained is the canonical one; this proves the first assertion.

Now suppose f finite and (f_*F, K_Y) bidualizing, since $Rf_* = f_*$ for f finite. We prove that the canonical morphism $F \rightarrow \underline{D}_X^2 F$ is an isomorphism. By Lemma 1.14(a) below, it suffices to show that the induced morphism

$$f_*F \longrightarrow f_*\underline{D}_X^2 F$$

is an isomorphism. But by applying (a)(i) twice, one convinces oneself that this morphism is precisely the canonical isomorphism

$$f_*F \longrightarrow \underline{D}_Y^2 f_*F,$$

which completes the proof. $\square$

Lemma 1.14. Let $f : X \rightarrow Y$ be a finite morphism of preschemes.

- a) For a morphism $u : F \rightarrow G$ in $D(X)$, u is an isomorphism if and only if f_*u is an isomorphism.
- b) For $F \in \text{ob } D(X)$, one has $F \in \text{ob } D_c(X)$ if and only if $f_*F \in \text{ob } D_c(Y)$.

Proof. For (a), since f_* is exact, one may restrict to the case where F and G are concentrated in degree zero. The assertion then follows immediately from the computation of the fibres of f_*F and f_*G , SGA A, Exposé VIII, 5.5.

For (b), again one may restrict to the case where F is concentrated in degree zero. Necessity is known, SGA A, Exposé IX, 2.14. We prove sufficiency: f_*F constructible implies F constructible.

The question is local upstairs, hence a fortiori downstairs, so we may suppose Y and hence X affine. By EGA IV, 17.16.6, there exists a finite family (Y_i) of affine subschemes of Y , pairwise disjoint and with union Y , such that, if $X_i = X \times_Y Y_i$, the induced morphism $f_i : X_i \rightarrow Y_i$ factors as

$$X_i \xrightarrow{u_i} X'_i \xrightarrow{v_i} Y_i,$$

where u_i is finite radicial and surjective and v_i is finite étale. Put $F|_{X_i} = F_i$. By the base-change theorem for finite morphisms, SGA A, Exposé VIII, 5.5, one has

$$f_{i*}(F_i) \xrightarrow{\sim} (f_*F)|_{Y_i},$$

and by SGA A, Exposé IX, 2.8, one is reduced to proving the assertion for (f_i, F_i) . Put $F'_i = u_{i*}(F_i)$, so that $f_{i*}(F_i) = v_{i*}(F'_i)$. It is clear from the definition of constructibility, respectively follows from SGA A, Exposé VIII, 1.1, that F_i , respectively F'_i , is constructible if and only if F'_i , respectively $v_{i*}(F'_i)$, is constructible. This proves the claim. $\square$

Corollary 1.15. Let $f : X \rightarrow Y$ be a separated quasi-finite morphism of noetherian preschemes. Let $K_Y \in \text{ob } D^+(Y)$, and put, as in Proposition 1.12,

$$K_X = R^!f(K_Y), \quad \underline{D}_X = \underline{D}_{K_X}, \quad \underline{D}_Y = \underline{D}_{K_Y}.$$

If K_Y is dualizing, then K_X is dualizing.

Proof. We already know, by Proposition 1.6, that if K_Y has finite quasi-injective dimension, then the same is true of K_X . We show that if K_Y satisfies condition (ii), respectively condition (iii), of Definition 1.7, then K_X satisfies the same condition.

By the Main Theorem, EGA IV, 8.12.6, f factors as $f = f'i$, where i is an open immersion and f' is finite. Since

$$R^!f \simeq R^!i R^!f',$$

one must treat separately the case of an open immersion and the case of a finite morphism. If f is an open immersion, the assertion is trivial, since a constructible sheaf on X is induced by a constructible sheaf on Y , and formation of $R\text{Hom}$ commutes with restriction to an open.

Suppose then that f is finite. It follows from Corollary 1.13 that if K_Y satisfies (iii), then K_X satisfies (iii), taking account of the finiteness theorem SGA A, Exposé IX, 2.14. It remains to prove that if K_Y satisfies (ii), then K_X satisfies (ii). Let $F \in \text{ob } D_c^-(X)$. We must show that $\underline{D}_X(F) \in \text{ob } D_c^+(X)$. By the finiteness theorem, $f_*F \in \text{ob } D_c^-(Y)$. Hence, by the hypothesis on K_Y ,

$$\underline{D}_Y f_*F \in \text{ob } D_c^+(Y).$$

But

$$\underline{D}_Y f_*F \xrightarrow{\sim} f_*\underline{D}_X(F)$$

by Proposition 1.12(a)(i), and Lemma 1.14(b) completes the proof. $\square$

2. Uniqueness of the dualizing complex

In this section we suppose that A is local, that is,

$$A = \mathbb{Z}/\ell^\nu \mathbb{Z},$$

where ℓ is prime.

Theorem 2.1, cf. [H] V.3.1. Let X be a connected locally noetherian prescheme, and let K be a dualizing complex on X in the sense of Definition 1.7. Let also $K' \in \text{ob } D_c^+(X)$. Then K' is dualizing if and only if there exist an invertible A_X -module L and an integer r , necessarily determined up to unique isomorphism, such that

$$K' \xrightarrow{\sim} K \otimes^L L[r].$$

Proof. Let L be an invertible A_X -module and r an integer, and put $L^\vee = \underline{\text{Hom}}(L, A_X)$. For $F \in \text{ob } D(X)$ and $G \in \text{ob } D^+(X)$ there are canonical isomorphisms

$$\text{R } \underline{\text{Hom}}(F \otimes^L L^\vee[-r], G) \xrightarrow{\sim} \text{R } \underline{\text{Hom}}(F, G \otimes^L L[r]) \xrightarrow{\sim} \text{R } \underline{\text{Hom}}(F, G) \otimes^L L[r]. \quad (*)$$

It follows immediately that, if $K' \simeq K \otimes^L L[r]$, then K' is dualizing.

Moreover, if we write $\underline{D}_X = \underline{D}$ and $\underline{D}_{K'} = \underline{D}'$, then for $F \in \text{ob } D(X)$ the isomorphism $(*)$ gives

$$\underline{D}'F \xrightarrow{\sim} \underline{D}F \otimes^L L^\vee[-r].$$

Since $\underline{D}K \simeq A_X$, one obtains

$$\underline{D}'K \xrightarrow{\sim} L^\vee[-r],$$

which shows that $L^\vee[r]$, hence L and r , are determined essentially uniquely by K' .

Conversely, suppose that K' is dualizing, and put

$$P = \underline{D}'K = \underline{D}'\underline{D}A_X = \text{R } \underline{\text{Hom}}(K, K').$$

We shall prove:

- (a) $K' \xrightarrow{\sim} K \otimes^L P$,
- (b) $P \xrightarrow{\sim} L[r]$ for an invertible A_X -module L .

This will complete the proof of the theorem.

First observe that, by Proposition 1.9(b)(ii), for $F \in \text{ob } D_c(X)$ there is an isomorphism

$$F \otimes^L P \xrightarrow{\sim} \underline{D}'\underline{D}F. \quad (**)$$

Applying $(**)$ to $F = K$ gives

$$K \otimes^L P \xrightarrow{\sim} \underline{D}'\underline{D}K \xrightarrow{\sim} \underline{D}'A_X = K',$$

which proves (a).

Applying $(**)$ to

$$F = P' = \underline{D}\underline{D}'A_X$$

gives

$$P' \otimes^L P = \underline{D}'\underline{D}\underline{D}\underline{D}'A_X \xrightarrow{\sim} \underline{D}'\underline{D}'A_X \xrightarrow{\sim} A_X.$$

Thus (b) follows from the following lemma. □

Lemma 2.2, cf. [H] V.3.3. Let X be a connected locally noetherian prescheme, and let $P, P' \in \text{ob } D_c^-(X)$ be such that

$$P \otimes^L P' \xrightarrow{\sim} A_X.$$

Then there exist an invertible A_X -module L and an integer r such that

$$P \xrightarrow{\sim} L[r], \quad P' \xrightarrow{\sim} L^\vee[-r].$$

Proof. The proof has two steps.

First suppose that $X = \text{Spec}(k)$, with k separably closed. One merely repeats the easy proof given in [H]. The essential point is this: by EGA 0_I, 5.4.3, if M and M' are finite-type A -modules such that

$$M \otimes M' \xrightarrow{\sim} A,$$

then M and M' are invertible and $M' \simeq M^\vee$. This is where the hypothesis that A is local is used.

We now pass to the general case. By the special case just treated, for every $x \in X$ there exists an integer $r(x)$ such that

$$P_{\bar{x}} \simeq A_{\bar{x}}[r(x)], \quad P'_{\bar{x}} \simeq A_{\bar{x}}[-r(x)].$$

Equivalently, $H^i(P_{\bar{x}}) = 0$, respectively $H^i(P'_{\bar{x}}) = 0$, except for $i = -r(x)$, respectively $i = r(x)$, and the corresponding nonzero cohomology modules are isomorphic to $A_{\bar{x}}$. It remains to show that the function $x \mapsto r(x)$ is locally constant.

Assume this for the moment. Since X is connected, r is constant, say equal to r_0 . Then

$$H^i(P) = 0 \ (i \neq -r_0), \quad H^i(P') = 0 \ (i \neq r_0),$$

and

$$H^{-r_0}(P) \otimes H^{r_0}(P') \xrightarrow{\sim} A.$$

Put $L = H^{-r_0}(P)$ and $L' = H^{r_0}(P')$. Let $u : \bar{x} \rightarrow \bar{y}$ be a specialization morphism, and let

$$u^* : L_{\bar{y}} \rightarrow L_{\bar{x}}, \quad u'^* : L'_{\bar{y}} \rightarrow L'_{\bar{x}}$$

be the corresponding specialization morphisms. There is a commutative square

$$\begin{array}{ccc} L_{\bar{y}} \otimes L'_{\bar{y}} & \xrightarrow{\sim} & A_{\bar{y}} \\ \downarrow u^* \otimes u'^* & & \downarrow \\ L_{\bar{x}} \otimes L'_{\bar{x}} & \xrightarrow{\sim} & A_{\bar{x}}. \end{array} \quad (*)$$

Choose a basis e , respectively f , of $L_{\bar{x}}$, respectively $L_{\bar{y}}$, and let e' , respectively f' , be the dual basis of $L'_{\bar{x}}$, respectively $L'_{\bar{y}}$, meaning that $e \otimes e'$ and $f \otimes f'$ map to 1. If $u^*(f) = \lambda e$ and $u'^*(f') = \lambda' e'$, the commutativity of $(*)$ gives $\lambda\lambda' = 1$. Hence u^* is an isomorphism, and u'^* is the contragredient isomorphism. By SGA A, Exposé IX, 2.13, it follows that L and L' are locally constant, hence of finite presentation as A_X -modules. Consequently, for every $x \in X$ the canonical map

$$\underline{\text{Hom}}(L, F)_{\bar{x}} \longrightarrow \text{Hom}(L_{\bar{x}}, F_{\bar{x}})$$

is an isomorphism for every A_X -module F . Since the geometric fibres of L are invertible, L itself is invertible. Similarly L' is invertible, and $L' \simeq L^\vee$. Hence

$$P \simeq L[-r_0], \quad P' \simeq L^\vee[r_0].$$

It remains only to prove that r is locally constant. For $m \in \mathbb{Z}$, the condition $r(x) = m$ is equivalent to

$$H^m(P')_{\bar{x}} \neq 0.$$

Thus, by SGA A, Exposé IX, 2.4, the function r is locally constructible. Therefore, to verify that it is locally constant, EGA IV, 1.10.1, allows one to suppose that X is local. Let x be its closed point and U the open complement.

We prove that r is constant. By induction on the dimension of X , we may already suppose that $r(y)$ is constant and equal to r_0 for $y \in U$; the point is to prove $r(x) = r_0$.

Assume the contrary, and derive a contradiction. By hypothesis

$$P|_U \xrightarrow{\sim} L[r_0]$$

with L an invertible A_U -module. After translating degrees we may suppose $r_0 = 0$. On the other hand $r(x) \neq 0$; write $r(x) = a > 0$. Let

$$i : U \rightarrow X, \quad j : x \rightarrow X$$

be the canonical inclusions. Consider the exact sequence

$$0 \longrightarrow i_!(P|_U) \longrightarrow P \longrightarrow j_*j^*P \longrightarrow 0. \quad (**)$$

By hypothesis the only nonzero cohomology objects of P are $H^0(P)$ and $H^{-a}(P)$, and

$$H^0(P) \xrightarrow{\sim} i_!L.$$

It follows that $(**)$ splits naturally, via the map $P \rightarrow H^0(P)$, hence

$$P \xrightarrow{\sim} i_!(P|_U) \oplus j_*j^*P.$$

Therefore

$$\begin{aligned} P \otimes^L P' &\xrightarrow{\sim} i_!(P|_U) \otimes^L P' \oplus j_*j^*P \otimes^L P' \\ &\xrightarrow{\sim} i_!(P|_U \otimes^L P'|_U) \oplus j_*(j^*P \otimes^L j^*P') \\ &\xrightarrow{\sim} i_!A_U \oplus j_*A_x, \end{aligned}$$

which is impossible, since $i_!A_U \oplus j_*A_x$ is plainly not isomorphic to A_X ; already the degree-zero cohomology sheaves are not isomorphic. This proves Lemma 2.2, and hence Theorem 2.1. $\square$

3. Existence of dualizing complexes

3.1. Preliminaries

As was said in the introduction, the object of this section is to prove that, on a “good” regular prescheme X , the sheaf A_X is dualizing, where n is assumed prime to the residual characteristics.

In fact, one would like this to hold on excellent regular preschemes of equal characteristic; but there is no doubt that the proof given below will extend to them once resolution of singularities and the purity theorem are available in that case.

We first present the various ingredients which will be used.

3.1.1. Singular locus. Let X be a prescheme, locally noetherian. We shall say that X satisfies condition *(reg)* if the following equivalent properties hold (EGA IV, 6.12.3):

- (i) for every sub-prescheme Y of X , the set $\text{Reg}(Y)$ of regular points of Y is open in Y ;
- (ii) for every closed integral sub-prescheme Y of X , the set $\text{Reg}(Y)$ contains a non-empty open subset of Y .

The condition *(reg)* is satisfied if X is locally of finite type over a complete noetherian local ring, for example over a field (EGA IV, 6.12.8), or over a Dedekind ring whose field of fractions has characteristic zero, for example over $\mathbb{Z}$ (EGA IV, 6.12.6); more generally, it is satisfied if X is excellent (EGA IV, 7.8.6).

3.1.2. Dévissage of constructible sheaves. Let X be a noetherian prescheme, and let C be a strictly full subcategory of the category $\text{Cons}(X)$ of constructible A_X -modules. Suppose that C is stable under direct factors and extensions. The following conditions are equivalent:

- (i) $C = \text{Cons}(X)$;
- (ii) C contains sheaves of the form $i_!(F)$, where $i : Y \rightarrow X$ is the immersion of an integral sub-prescheme and F is locally constant on Y ;
- (iii) C contains sheaves of the form $f_*i_!(A'_U)$, where $i : U \rightarrow Y$ is an open subset of an integral prescheme, $f : Y \rightarrow X$ is a finite morphism such that the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$, for ℓ a prime divisor of n .

If X satisfies *(reg)* (3.1.1), these conditions are again equivalent to the following:

- (i bis) the analogue of (ii), with Y moreover regular;
- (ii bis) the analogue of (iii), with U regular.

If, in addition, X is affine, then in (ii), respectively in (ii bis), one may suppose Y closed in an open subset of the form $D(f)$, with $f \in \Gamma(X, \mathcal{O}_X)$.

Proof. The equivalence of (i)–(iii) was proved in SGA A, Exposé IX, 2.5 and 5.8. The remaining assertions follow by noetherian induction. $\square$

3.1.3. Cohomological dimension. Let X be a locally noetherian prescheme. We denote by $(\text{cdloc})_n$ the following condition. For every sub-prescheme Y of X , every geometric point $\bar{y}$ of Y , and every

$$f \in \Gamma(\tilde{Y}(\bar{y}), \mathcal{O}_{\tilde{Y}, \bar{y}}),$$

where $\tilde{Y}(\bar{y})$ denotes the strict localization of Y at $\bar{y}$ (SGA A, Exposé VIII, 7.1), one has

$$\text{cd}_{\mathbf{n}}(\tilde{Y}(\bar{y}) - V(f)) \leq \dim \mathcal{O}_{Y, \bar{y}} \quad (= \dim \mathcal{O}_{Y, y} \text{ by EGA IV, 6.1.3}),$$

where $\mathbf{n}$ is the set of prime divisors of n .

The condition $(\text{cdloc})_n$ is satisfied if X is locally of finite type over a field (SGA A, Exposé XIV, 3.5), or is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 6.2).⁴

3.1.4. Absolute cohomological purity. Let X be a regular prescheme, and let Y be a regular closed sub-prescheme of X , of codimension d at every point. One says that the pair (X, Y) satisfies the absolute cohomological purity theorem, relative to n , if

$$\underline{H}_Y^i(A_X) = 0 \quad (i \neq 2d),$$

and if the “local fundamental class” homomorphism (SGA 4^{1/2}, Cycle 2.2)

$$A_Y \longrightarrow \underline{H}_Y^{2d}((\mu_n)_X^{\otimes d})$$

is an isomorphism.

One says that the absolute cohomological purity theorem is true on X if every pair (U, Y) as above satisfies it, where U is an open subset of X .

Absolute purity is true on X if X is smooth over a perfect field of characteristic prime to n (SGA A, Exposé XVI, 3.9), or if X is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 3.2), or if X is regular of dimension 1 (cf. 5.1).⁵

3.1.5. Hironaka-style resolution of singularities.

- a) Let X be a regular prescheme, and let $D \geq 0$ be a divisor on X . One says that D has *strict normal crossings* if there exists a finite family $(f_i)_{i \in I}$ of elements of $\Gamma^*(X, \mathcal{O}_X)$ such that

$$D = \sum_{i \in I} \text{Div}(f_i)$$

(EGA IV, 21), and such that, for every $x \in \text{Supp}(D)$, the germs $(f_i)_x$ which lie in $\mathfrak{m}_x$ form part of a regular system of parameters. One says that D has *normal crossings* if, locally for the étale topology, D has strict normal crossings.

The standard example of a normal crossings divisor is furnished by a locally finite union of coordinate hyperplanes in affine space.

- b) Let X be a locally noetherian prescheme. We shall say that X is *strongly desingularizable* if the following condition is satisfied: for every finite morphism $f : Y \rightarrow X$, where Y is integral and its generic point is separable over its image, and every regular open subset U of Y , there exists a proper birational morphism $h : Y' \rightarrow Y$, with Y' regular, such that h induces an isomorphism from $U' = h^{-1}(U)$ onto U , and $Y' - U'$ is a normal crossings divisor.

Here are cases where X is strongly desingularizable:

- a) X is excellent of characteristic zero;
- b) X is locally noetherian, regular, and of dimension ≤ 1 ;

⁴Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.

⁵Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.

c) X is of finite type over $\mathbb{Z}$ or over a perfect field, and of dimension ≤ 2 .⁶

3.1.6. Cohomological finiteness. Let $f : X \rightarrow Y$ be a morphism of preschemes, and let $F \in \text{ob } D_c^+(X)$. One has $Rf_*(F) \in \text{ob } D_c^+(Y)$ in the following cases:

- a) f is proper and of finite presentation (SGA A, Exposé XIV, 1.1);
- b) “when resolution of singularities and the purity theorem are available”, in particular:
 - (i) if X and Y are locally of finite type over a field k of characteristic zero and f is a k -morphism of finite type (SGA A, Exposé XVI, 5.1);
 - (ii) if Y is excellent of characteristic zero and f is of finite type (SGA A, Exposé XIX, 5.1);
 - (iii) if X is locally of finite type over a field and $\dim(X) \leq 2$ (SGA A, Exposé XIX, 5.1).

3.2. The quasi-injective dimension of A_X

Proposition 3.2.1. Let X be a regular prescheme whose residual characteristics are prime to n . Suppose that X satisfies (reg) (3.1.1) and (cdloc) $_n$ (3.1.3), and that absolute purity is true on X (3.1.4). Then, for every constructible A_X -module F and every $x \in X$, one has

$$\underline{\text{Ext}}^i(F, A_X)_x = 0 \quad \text{for } i > 2 \dim \mathcal{O}_{X,x}. \quad (*)$$

Proof. The question being local, we may suppose X affine. By dévissage (3.1.2), it is enough to verify (*) for $F = i_!(G)$, where $i : Y \rightarrow X$ is the immersion of a regular integral sub-prescheme, closed in an open subset $D(f)$, $f \in A(X)$, and G is a locally constant constructible sheaf on Y . We must compute

$$\underline{\text{Ext}}^q(F, A_X) = H^q(R \underline{\text{Hom}}(F, A_X)).$$

The projection formula for i (SGA A, Exposé XVIII) gives

$$R \underline{\text{Hom}}(i_! G, A_X) \xrightarrow{\sim} Ri_* R \underline{\text{Hom}}(G, R^! i(A_X)).$$

By purity,

$$R^! i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes d}[-2d],$$

where $d = \text{codim}(Y, X)$. Put $T = (\mu_n)_Y^{\otimes d}$. Since G is locally constant constructible, for every geometric point $\bar{y}$ of Y and every p one has

$$\underline{\text{Ext}}^p(G, T)_{\bar{y}} \xrightarrow{\sim} \text{Ext}^p(G_{\bar{y}}, T_{\bar{y}}).$$

But $T_{\bar{y}}$ is injective, being isomorphic to $A = \mathbb{Z}/n\mathbb{Z}$. Hence $\underline{\text{Ext}}^p(G, T) = 0$ for $p \neq 0$, and

$$R \underline{\text{Hom}}(G, T) \xrightarrow{\sim} \underline{\text{Hom}}(G, T).$$

⁶Added in 1977: more generally, if X is excellent noetherian of dimension ≤ 2 , by H. Hironaka, *Desingularization of excellent surfaces*, notes by B. Bennett, Bowdoin College, 1967, to be supplemented if necessary by H. Hironaka, *Certain numerical characters of singularities*, J. Math. Kyoto Univ. 10 (1970), 151–187.

Thus

$$\mathrm{R} \underline{\mathrm{Hom}}(F, A_X) \xrightarrow{\sim} \mathrm{R} i_*(\mathcal{H})[-2d], \quad \mathcal{H} = \underline{\mathrm{Hom}}(G, T).$$

Let Z be the schematic closure of Y in X (EGA I, 9.5.10). Then $Y = Z \cap D(f)$ (EGA 0_I, 2.1.6). Denote by $j : Y \rightarrow Z$ and $k : Z \rightarrow X$ the canonical immersions. We have $\mathrm{R} i_* = k_* \mathrm{R} j_*$, so we are reduced to computing $\mathrm{R} j_*(\mathcal{H})$. If $\bar{x}$ is a geometric point of Z , let $\tilde{Z} = \mathrm{Spec}(\mathcal{O}_{Z, \bar{x}})$ be the strict localization, let $\tilde{f}$ be the image of f in $\mathcal{O}_{Z, \bar{x}}$, let $\tilde{Y} = \tilde{Z} \times_Z Y = \tilde{Z} - V(\tilde{f})$, and let $\tilde{\mathcal{H}}$ be the inverse image of $\mathcal{H}$ on $\tilde{Y}$. By SGA A, Exposé VIII, 5.2,

$$\mathrm{R}^q j_*(\mathcal{H})_{\bar{x}} \xrightarrow{\sim} \mathrm{H}^q(\tilde{Y}, \tilde{\mathcal{H}}),$$

hence

$$\underline{\mathrm{Ext}}^q(F, A_X)_{\bar{x}} \xrightarrow{\sim} \mathrm{H}^{q-2d}(\tilde{Y}, \tilde{\mathcal{H}}).$$

The condition $(\mathrm{cdloc})_n$ implies that this group is zero for

$$q - 2d > \dim \mathcal{O}_{Z, \bar{x}}.$$

Using

$$d = \mathrm{codim}_{\bar{x}}(Z, X) \leq \dim \mathcal{O}_{X, \bar{x}}$$

(EGA IV, 5.1.3) and

$$\dim \mathcal{O}_{Z, \bar{x}} = \dim \mathcal{O}_{X, \bar{x}} - \mathrm{codim}_{\bar{x}}(Z, X)$$

(EGA 0_{IV}, 16.5.12 and EGA IV, 5.1.9), one obtains

$$\underline{\mathrm{Ext}}^q(F, A_X)_x = 0 \quad (q > 2 \dim \mathcal{O}_{X, x}),$$

as required. $\square$

Remark 3.2.2. The bound obtained in 3.2.1 is the best possible. Indeed, let x be a point of X , and let Y be the closed integral sub-prescheme of X whose generic point is x . Since X satisfies (reg), there exists a regular open subset U of Y containing x . By purity,

$$\underline{\mathrm{Ext}}^{2d}(A_U, A_X)|_U = (\mu_n)_U^{\otimes d},$$

where $d = \dim \mathcal{O}_{X, x} = \mathrm{codim}(Y, X)$.

Corollary 3.2.3. Let X be a regular prescheme of dimension d . If X is smooth over a field k of characteristic prime to n , or if X is excellent of characteristic zero, then X satisfies the conclusion of 3.2.1; consequently

$$\dim . q . \mathrm{inj}(A_X) \leq 2d.$$

Proof. If X is excellent of characteristic zero, or if k is perfect of characteristic prime to n and X is smooth over k , then the hypotheses of 3.2.1 are satisfied by 3.1.1, 3.1.3, and 3.1.4. The conclusion is still valid when X is smooth over an arbitrary field k , as follows immediately after the base change $\mathrm{Spec}(k') \rightarrow \mathrm{Spec}(k)$, where k' is a perfect closure of k (SGA A, Exposé VIII, 1.1). $\square$

Lemma 3.2.4. Let X be a prescheme satisfying the conclusion of 3.2.1 and of dimension $\leq d$. If there exists an integer N such that X has cohomological dimension $\leq N$, then

$$\dim . \mathrm{inj}(A_X) \leq 2d + N.$$

If, moreover, for every closed subset Y of X , the implication

$$\mathrm{codim}(Y, X) \geq p \quad \Rightarrow \quad \mathrm{cd}_{\mathbf{n}}(Y) \leq N - 2p$$

holds, then

$$\dim \cdot \mathrm{inj}(A_X) \leq N.$$

Proof. The first assertion follows immediately from 1.5. For the second, one argues as in 1.5. We are reduced to proving, for F constructible on X , the relation

$$\mathrm{Ext}^i(X; F, A_X) = 0 \quad (i > N).$$

For this, inspect the spectral sequence

$$E_2^{pq} = H^p(X; \underline{\mathrm{Ext}}^q(F, A_X)) \Rightarrow \mathrm{Ext}^*(X; F, A_X).$$

By 3.2.1, the sheaves $\underline{\mathrm{Ext}}^{2i}(F, A_X)$ and $\underline{\mathrm{Ext}}^{2i-1}(F, A_X)$ are concentrated in codimension $\geq i$. Therefore $E_2^{p, 2i} = E_2^{p, 2i-1} = 0$ for $p > N - 2i$, by the hypothesis. Hence $E_2^{pq} = 0$ for $p + q > N$, and the assertion follows. $\square$

Corollary 3.2.5. If X is a smooth prescheme of dimension d over a separably closed field k of characteristic prime to n , then

$$\dim \cdot \mathrm{inj}(A_X) = 2d.$$

Proof. By 3.2.3 and SGA A, Exposé X, 4.2, we are in the situation of 3.2.4. The hypotheses of SGA A, Exposé X, 4.2 are satisfied by SGA A, Exposé X, 2.1 and 3.2; hence $\dim \cdot \mathrm{inj}(A_X) \leq 2d$. The equality follows from purity: if x is a closed point of codimension d in X , then

$$\mathrm{Ext}^{2d}(X, A_x, A_X) = H_x^{2d}(A_X) \xrightarrow{\sim} (\mu_n)^{\otimes d}(k) \neq 0.$$

$\square$

3.3. Constructibility of the sheaves $\underline{\mathrm{Ext}}^i(F, A_X)$

Proposition 3.3.1. Let X be a prescheme.

a) The following conditions are equivalent:

(i) for $F \in \mathrm{ob} D_c^-(X)$ and $G \in \mathrm{ob} D_c^+(X)$, one has

$$\mathrm{R} \underline{\mathrm{Hom}}(F, G) \in \mathrm{ob} D_c^+(X);$$

(i bis) for every pair (F, G) of constructible A_X -modules, one has $\mathrm{R} \underline{\mathrm{Hom}}(F, G) \in \mathrm{ob} D_c^+(X)$, i.e. the sheaves $\underline{\mathrm{Ext}}^i(F, G)$ are constructible for all i ;

(ii) for every open immersion $i : U \rightarrow X$ and every constructible A_U -module F , the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p ;

(ii bis) for every immersion $i : Y \rightarrow X$ and every constructible A_Y -module F , the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p ;

- (ii ter) for every immersion $i : Y \rightarrow X$ and every locally constant constructible A_Y -module F , the sheaves $R^p i_*(F)$ are constructible for all p ;
- (iii) for every finite morphism, respectively every closed immersion, $f : Y \rightarrow X$, and every $F \in \text{ob } D_c^+(X)$, one has $R^! f(F) \in \text{ob } D_c^+(Y)$.

b) These conditions are satisfied in each of the following two cases:

- (i) $\dim(X) \leq 1$;
- (ii) resolution of singularities and purity are available on X in the sense of 3.1.6(b).

Proof. The equivalence of (i) and (i bis) follows from the standard spectral sequence passing from sheaf-Ext to hyper-Ext.

(i bis) $\Rightarrow$ (ii). Let $\bar{F}$ be a constructible A_X -module extending F , for example $\bar{F} = i_! F$. By duality (SGA A, Exposé XVIII, 3.1.10),

$$Ri_* F = Ri_* R \underline{\text{Hom}}(A_U, i^! \bar{F}) \xrightarrow{\sim} R \underline{\text{Hom}}(i_! A_U, \bar{F}),$$

and this proves the claim.

(ii) $\Rightarrow$ (ii bis). Factor i as a closed immersion followed by an open immersion.

(ii bis) $\Rightarrow$ (i bis). The question is local, so we may suppose X affine. By the usual dévissage (SGA A, Exposé IX, 2.5), it is enough to verify (ii) for $F = i_! M$, where $i : Y \rightarrow X$ is the immersion of a sub-prescheme and M is a locally constant constructible A_Y -module. By the projection formula (SGA A, Exposé XVIII),

$$R \underline{\text{Hom}}(i_! M, G) \xrightarrow{\sim} Ri_* R \underline{\text{Hom}}(M, Ri^! G),$$

and by a spectral sequence we are reduced to proving that

$$R \underline{\text{Hom}}(M, Ri^! G) \in \text{ob } D_c^+(Y).$$

First prove $Ri^! G \in \text{ob } D_c^+(Y)$. After factoring i , we may suppose that Y is closed. Let $j : U = X - Y$ be the complementary open immersion. Applying $R \underline{\text{Hom}}(_, G)$ to

$$0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0$$

gives a distinguished triangle. Since $R \underline{\text{Hom}}(A_X, G) = G$, and

$$R \underline{\text{Hom}}(A_{U,X}, G) \xrightarrow{\sim} Rj_* j^! G$$

belongs to $D_c^+(X)$ by hypothesis, it follows that

$$R \underline{\text{Hom}}(A_Y, G) \xrightarrow{\sim} i_* Ri^! G$$

belongs to $D_c^+(X)$. Resolving M locally on Y , for the étale topology, by free A_Y -modules, another dévissage gives

$$R \underline{\text{Hom}}(M, Ri^! G) \in \text{ob } D_c^+(Y).$$

This proves the implication.

(ii bis) $\Rightarrow$ (ii ter) is trivial.

(ii ter) $\Rightarrow$ (ii). Let $i : U \rightarrow X$ be an open immersion and let F be a constructible A_U -module. We must prove $Ri_*F \in \text{ob } D_c^+(X)$. The question being local, suppose X noetherian and argue by noetherian induction on X . We may suppose $U \neq \emptyset$. Then there exists a non-empty open subset $j : V \rightarrow U$ such that j^*F is locally constant (SGA A, Exposé IX, 2.4). Let M be the mapping cylinder of

$$F \longrightarrow Rj_*j^*F.$$

Applying Ri_* gives a distinguished triangle

$$Ri_*F \longrightarrow Ri_*Rj_*j^*F \longrightarrow Ri_*M \xrightarrow{[1]}.$$

Condition (ii ter) implies

$$Ri_*Rj_*j^*F \xrightarrow{\sim} R(ij)_*j^*F \in \text{ob } D_c^+(X).$$

Consequently,

$$Rj_*j^*F \xrightarrow{\sim} i^*R(ij)_*j^*F \in \text{ob } D_c^+(U).$$

Thus $M \in \text{ob } D_c^+(U)$, since $F \in \text{ob } D_c^+(U)$. Moreover, the cohomology of M is supported on a closed subset Z of U distinct from U . Applying the induction hypothesis to $U \cap \overline{Z} \rightarrow \overline{Z}$, we deduce $Ri_*M \in \text{ob } D_c^+(X)$, and hence $Ri_*F \in \text{ob } D_c^+(X)$.

The equivalence of (i)–(ii ter) with condition (iii) is left to the reader; use 1.14.

For b)(i), normalization reduces one to the case where X is regular, and one uses purity, which will be proved in 5.1. For b)(ii), see 3.1.6(b). $\square$

3.4. The biduality theorem on regular preschemes

Theorem 3.4.1, cf. [H], V, 2.2. Let X be a regular locally noetherian prescheme of finite dimension. Suppose that X is strongly desingularizable (3.1.5), satisfies (reg) (3.1.1) and (cdloc) $_n$ (3.1.3), and that absolute purity (3.1.4) is true on every prescheme smooth over X . Then A_X is dualizing.

Proof. We must prove that A_X satisfies conditions (i)–(iii) of 1.6. Condition (i) follows from 3.2.1. Condition (ii) follows from 3.3.1(b). It remains to prove condition (iii). Thus, by 1.7, for every constructible A_X -module F we must prove that the canonical homomorphism

$$F \longrightarrow \underline{D}_X \underline{D}_X F, \quad \underline{D}_X = R\underline{\text{Hom}}(\ , A_X),$$

is an isomorphism. The question being local, suppose X noetherian. By dévissage (3.1.2, iii bis), it is enough to treat sheaves

$$F = f_*i_!(A'_U),$$

where $i : U \rightarrow Y$ is the inclusion of a regular open subset of an integral prescheme, $f : Y \rightarrow X$ is finite and the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$ for a prime divisor ℓ of n .

Because X is strongly desingularizable, there is a proper birational morphism $h : Y' \rightarrow Y$, with Y' regular, inducing an isomorphism from $U' = h^{-1}U$ to U , and such that $Y' - U'$ is a normal crossings divisor. By base change (SGA A, Exposé XVII),

$$i_!(A'_U) \xrightarrow{\sim} Rh_*i'_!(A'_{U'}),$$

where $i' : U' \rightarrow Y'$ is the inclusion. Hence, with $g = fh$,

$$f_* i_!(A'_U) \xrightarrow{\sim} f_* R h_* i'_!(A'_{U'}) \xrightarrow{\sim} R g_* i'_!(A'_{U'}).$$

By 1.11, it suffices to prove that the pair

$$(i'_!(A'_{U'}), R^! g(A_X))$$

is bidualizing. Locally immersing Y' in a prescheme X' smooth over X , and applying purity on X' , one sees that $R^! g(A_X)$ is locally isomorphic to $A_{Y'}$, up to a translation of degrees. We are therefore reduced to the following special case.

Lemma 3.4.2. Let X be a locally noetherian regular prescheme on which absolute purity is true. Let $Y = Y_1 \cup \cdots \cup Y_p$ be a strict normal crossings divisor on X , and put $U = X - Y$. Then the pair $(A'_{U,X}, A_X)$ is bidualizing.

Proof. By the exact sequence

$$0 \longrightarrow A'_{U,X} \longrightarrow A'_X \longrightarrow A'_Y \longrightarrow 0,$$

it is equivalent to verify that (A'_Y, A_X) is bidualizing. We argue by induction on p .

If $p = 1$, let $i : Y \rightarrow X$ be the immersion. By 1.11, it is enough to verify that

$$(A'_Y, R^! i(A_X))$$

is bidualizing. By purity,

$$R^! i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes 1}[-2].$$

Thus the canonical homomorphism $A'_Y \rightarrow \underline{D}_Y \underline{D}_Y(A'_Y)$ is an isomorphism, as is checked trivially fibre by fibre, using Pontryagin duality for A -modules.

Suppose $p \geq 2$, and put $Y' = Y_1 \cup \cdots \cup Y_{p-1}$. We have an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_Y \longrightarrow A'_{Y'} \longrightarrow 0.$$

By the induction hypothesis, $(A'_{Y'}, A_X)$ is bidualizing. By the five lemma, we are reduced to biduality for $(A'_{(Y-Y'),X}, A_X)$. But again there is an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_{Y_p} \longrightarrow A'_{Y_p \cap Y'} \longrightarrow 0.$$

Since (A'_{Y_p}, A_X) is bidualizing, it remains to prove the same for $(A'_{Y_p \cap Y'}, A_X)$. Applying 1.11 again and using purity, we are finally reduced to biduality for this pair on Y_p , where the induction hypothesis applies. $\square$

This completes the proof of Theorem 3.4.1.

Corollary 3.4.3. Let S be an excellent noetherian regular prescheme of characteristic zero. Then A_S is dualizing. Moreover, if $f : X \rightarrow S$ is a morphism of finite type, then $R^! f(A_S)$ is dualizing.

Proof. The first assertion follows because S satisfies the hypotheses of 3.4.1, as recalled in 3.1. For the second, one must first verify that $R^! f(A_S)$ has finite quasi-injective dimension. Cover S by finitely many affine open subsets S_i such that $X_i = f^{-1}(S_i)$ is a finite union of affine open subsets

X_{ij} immersed as closed subschemes in preschemes X'_{ij} smooth of finite dimension over S_i . From 3.2.3 and 1.6 it follows that

$$\dim . q. inj(R^! f(A_S)) \leq 2 \sup \dim X'_{ij}.$$

It remains to verify conditions (ii) and (iii) of 1.7. Since these are local on X for the étale topology, we may suppose that f factors as $f = f' i$, where $f' : X' \rightarrow S$ is smooth of relative dimension d , and $i : X \rightarrow X'$ is a closed immersion. Then

$$R^! f(A_S) \simeq R^! i R^! f'(A_S).$$

But $R^! f'(A_S)$ is locally isomorphic to $A_{X'}$, up to the translation by $2d$, hence is dualizing by the first assertion. By 1.15, $R^! f(A_S)$ is dualizing, and in particular satisfies conditions (ii) and (iii) of 1.7. $\square$

Corollary 3.4.4. Let k be a field, and let $f : X \rightarrow S = \text{Spec}(k)$ be a morphism locally of finite type, with $\dim(X) \leq 2$. Then $R^! f(A_S)$ is dualizing.

Proof. Replacing k by its perfect closure, which is harmless for the étale topology (SGA A, Exposé VIII, 1.1), we may suppose k perfect. Cover X by affine open subsets X_i of finite type over k . Since $\dim(X_i) \leq 2$, the normalization lemma makes X_i finite over a scheme X'_i smooth over S and of dimension ≤ 2 . By 1.15 this reduces us to the case where f is smooth, hence locally

$$f^!(A_S) \simeq A_X[2d].$$

Thus it remains to show that A_X is dualizing in this case; this is exactly 3.4.1, since the hypotheses are satisfied (see 3.1). $\square$

Remark 3.4.5. We shall see below that if X is regular of dimension ≤ 1 , then A_X is dualizing.